

MAHAMUKABLA 26

COMPLETE HIGH-YIELD REVISION NOTES (JEE / NEET)

1. Electrostatics: Electric Charge and Coulomb's Law

Electric charge is an intrinsic property of elementary particles that gives rise to electric force. Quantization of charge implies that charge exists only in discrete integral multiples of elementary charge e .

$$Q = \pm ne \quad \text{where } e = 1.6 \times 10^{-19} \text{ C}$$

Properties of Charges:

- **Conservation:** The total charge of an isolated system remains constant.
- **Invariance:** The magnitude of charge is independent of its speed.

Coulomb's Law:

The electrostatic force between two stationary point charges is directly proportional to the product of their magnitudes and inversely proportional to the square of the distance between them.

$$F = k \cdot (|q_1 q_2| / r^2) = (1 / 4\pi\epsilon_0) \cdot (|q_1 q_2| / r^2)$$

Where ϵ_0 is the permittivity of free space = $8.854 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$, and $k \approx 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.

In a dielectric medium of relative permittivity ϵ_r (or dielectric constant K):

$$F_{\text{medium}} = F_{\text{vacuum}} / K = (1 / 4\pi\epsilon_0 K) \cdot (|q_1 q_2| / r^2)$$

2. Electric Field Intensity (E)

Electric field intensity at any point is defined as the electrostatic force experienced by a unit positive test charge placed at that point.

$$E = F / q_0 = (1 / 4\pi\epsilon_0) \cdot (q / r^2) \hat{r}$$

Principle of Superposition:

When multiple charges are interacting, the total force or electric field on a given charge is the vector sum of individual forces or fields exerted by all other charges.

$$E_{\text{total}} = E_1 + E_2 + E_3 + \dots + E_n$$

Electric Field due to Continuous Charge Distributions:

- **Linear Charge Density (λ):** $\lambda = q / L$ (Coulomb/meter)
- **Surface Charge Density (σ):** $\sigma = q / A$ (Coulomb/meter²)
- **Volume Charge Density (ρ):** $\rho = q / V$ (Coulomb/meter³)

3. Electric Dipole

An electric dipole consists of a pair of equal and opposite point charges separated by a small distance $2a$.

$$\text{Dipole Moment: } p = q \times 2a \text{ (Directed from } -q \text{ to } +q)$$

Electric Field due to Dipole:

- **Axial Point (at distance r from center):** For $r \gg a$:

$$E_{\text{axial}} = (1 / 4\pi\epsilon_0) \cdot (2p / r^3)$$

- **Equatorial Point (at distance r from center):** For $r \gg a$:

$$E_{\text{equatorial}} = (1 / 4\pi\epsilon_0) \cdot (p / r^3)$$

Dipole in Uniform External Electric Field:

- Torque: $\tau = p \times E$ or $\tau = pE \sin\theta$
- Potential Energy stored: $U = -p \cdot E = -pE \cos\theta$
- Work done in rotating from θ_1 to θ_2 : $W = pE(\cos\theta_1 - \cos\theta_2)$

4. Electric Flux and Gauss's Law

Electric flux measures the number of electric field lines passing through a given surface.

$$\Phi = \int \mathbf{E} \cdot d\mathbf{A} = E A \cos\theta$$

Gauss's Law Statement:

The total net electric flux through any closed surface (Gaussian surface) is equal to $1/\epsilon_0$ times the net charge enclosed by that closed surface.

$$\Phi_{\text{net}} = \oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enclosed}} / \epsilon_0$$

Key Applications:

1. Infinitely Long Straight Wire: $E = \lambda / (2\pi\epsilon_0 r)$
2. Infinite Thin Plane Sheet of Charge: $E = \sigma / (2\epsilon_0)$
3. Thin Spherical Shell (Radius R):
 - Outside ($r > R$): $E = q / (4\pi\epsilon_0 r^2)$
 - On Surface ($r = R$): $E = q / (4\pi\epsilon_0 R^2)$
 - Inside ($r < R$): $E = 0$

5. Electrostatic Potential and Potential Energy

Electric potential at a point is the amount of work done in bringing a unit positive charge from infinity to that point against the electrostatic forces.

$$V = W / q_0 = (1 / 4\pi\epsilon_0) \cdot (q / r)$$

Relation between Electric Field and Potential:

$$\mathbf{E} = -dV / d\mathbf{r} \quad \text{or} \quad \mathbf{E} = -(\partial V / \partial x \hat{i} + \partial V / \partial y \hat{j} + \partial V / \partial z \hat{k})$$

Electrostatic Potential Energy of System:

- For two point charges: $U = (1 / 4\pi\epsilon_0) \cdot (q_1 q_2 / r)$

- For three charges: $U = (1 / 4\pi\epsilon_0) \cdot (q_1q_2/r_{12} + q_2q_3/r_{23} + q_1q_3/r_{13})$

6. Capacitors and Capacitance

Capacitance is the ability of a system to store charge and electrical energy.

$$Q = C V \quad \Rightarrow \quad C = Q / V \text{ (Farad, F)}$$

Parallel Plate Capacitor:

$$C = \epsilon_0 A / d \quad (\text{With vacuum between plates})$$

If a dielectric medium of constant K fills the entire space:

$$C' = K \cdot C = K\epsilon_0 A / d$$

Combinations of Capacitors:

- Series: $1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3 + \dots$
- Parallel: $C_{eq} = C_1 + C_2 + C_3 + \dots$

Energy Stored in a Capacitor:

$$U = (1/2) Q V = (1/2) C V^2 = Q^2 / (2C)$$

7. Current Electricity: Electric Current and Ohm's Law

Electric current is the rate of flow of charge through any cross-section of a conductor.

$$I = dq / dt \quad \text{Current Density: } J = I / A$$

Drift Velocity (v_d):

The average velocity attained by free electrons in a conductor under the influence of an external electric field.

$$v_d = (e E \tau) / m$$

Where τ is the relaxation time and m is the mass of electron.

Relation between Current and Drift Velocity:

$$I = n e A v_d$$

Ohm's Law:

$$V = I R \quad \text{where Resistance } R = \rho(L / A)$$

Conductivity (σ) is the inverse of resistivity (ρ): $\sigma = 1 / \rho = n e^2 \tau / m$.

8. Kirchhoff's Laws and DC Circuits

Kirchhoff's First Law (Current Law - KCL):

The algebraic sum of currents meeting at any junction in an electrical circuit is zero. (Based on **Conservation of Charge**).

$$\sum I = 0$$

Kirchhoff's Second Law (Voltage Law - KVL):

The algebraic sum of changes in potential around any closed loop is zero. (Based on **Conservation of Energy**).

$$\sum \Delta V = 0 \quad \text{or} \quad \sum E = \sum I R$$

Cell EMF and Internal Resistance:

Terminal Potential Difference: $V = E - I r$ (during discharging), and $V = E + I r$ (during charging).

$$\text{Internal Resistance: } r = R \cdot (E/V - 1)$$

9. Electrical Measuring Instruments

Wheatstone Bridge:

Balanced condition when no current flows through the central galvanometer:

$$P / Q = R / S$$

Potentiometer:

Works on the principle that the potential drop across any portion of a wire of uniform cross-section is directly proportional to its length ($V \propto L$), provided a constant current flows.

$$\text{Potential Gradient: } k = V / L$$

- Comparison of EMF of two cells: $E_1 / E_2 = l_1 / l_2$
- Internal Resistance of a cell: $r = l_1 / l_2 - 1 \cdot R_{ext}$

10. Chemistry: Mole Concept Basics

A mole is defined as the amount of substance containing as many elementary entities as there are atoms in exactly 12 grams of Carbon-12.

$$\text{Avogadro's Number (N}_A\text{)} = 6.022 \times 10^{23} \text{ particles / mol}$$

Fundamental Mole Formulae:

1. In terms of mass: $\text{Moles} = \text{Given Mass (g)} / \text{Molar Mass (g/mol)}$
2. In terms of particles: $\text{Moles} = \text{Number of particles} / N_A$
3. In terms of gas volume at STP (Standard Temp & Pressure):

$$\text{Moles} = \text{Volume of gas at STP (in Liters)} / 22.4 \text{ L}$$

(Note: Modern IUPAC STP volume is 22.7 L at 1 bar, but standard exam problems widely use 22.4 L).

11. Concentration Expressions in Solutions

Concentration tells us how much solute is dissolved in a specific amount of solvent or solution.

Term	Formula	Temperature Dependence
Molarity (M)	Moles of solute / Volume of Solution (L)	Dependent (Changes with T)
Molality (m)	Moles of solute / Mass of Solvent (kg)	Independent
Mole Fraction (X)	$n_A / (n_A + n_B)$	Independent
Normality (N)	Gram equivalents / Volume of Solution (L)	Dependent

Relationship: Normality = Molarity $\times$ n-factor

12. Empirical Formula and Stoichiometric Calculations

Empirical vs Molecular Formula:

- **Empirical Formula:** Represents the simplest whole-number ratio of atoms of various elements present in a compound.
- **Molecular Formula:** Represents the actual number of atoms of each element present in a molecule.

Molecular Formula = $n \times$ Empirical Formula where $n = \text{Molar Mass} / \text{Empirical Formula Mass}$

Limiting Reagent:

The reactant that is completely consumed first in a chemical reaction, bounding the maximum amount of product formed. To find it, divide the initial moles of each reactant by its respective stoichiometric coefficient; the smallest value indicates the limiting reagent.

13. Classification of Elements & Periodic Properties

Modern Periodic Law states that the physical and chemical properties of elements are periodic functions of their **atomic numbers**.

Division into Blocks:

- **s-block:** Group 1 (Alkali metals) & Group 2 (Alkaline earth metals). General outer electronic configuration: ns^{1-2} .
- **p-block:** Groups 13 to 18. General configuration: $ns^2 np^{1-6}$. Together with s-block, called Representative Elements.
- **d-block:** Groups 3 to 12 (Transition Elements). Configuration: $(n-1)d^{1-10} ns^{0-2}$.
- **f-block:** Lanthanides and Actinides (Inner Transition Elements). Configuration: $(n-2)f^{1-14} (n-1)d^{0-1} ns^2$.

14. Periodic Trends: Atomic Size & Ionization Enthalpy

Atomic Radius Trend:

- **Across a Period:** Decreases because Effective Nuclear Charge (Z_{eff}) increases, pulling the electron cloud closer.
- **Down a Group:** Increases because the number of principal energy shells increases, dominating over increased nuclear charge.

Ionization Enthalpy (IE):

The minimum energy required to remove the most loosely bound electron from an isolated gaseous atom in its ground state.



- Across a Period: IE generally increases due to decreasing size and increasing Z_{eff} .
- Down a Group: IE decreases as atomic size increases, making outer electrons easier to detach.
- *Exceptions:* Half-filled and fully-filled configurations (e.g., IE_1 of $N > O$ and IE_1 of $Be > B$).

15. Electron Gain Enthalpy and Electronegativity

Electron Gain Enthalpy ($\Delta_{eg}H$):

The enthalpy change that occurs when an electron is added to an isolated gaseous atom.

- Halogens have highly negative electron gain enthalpies because they require only one electron to attain stable noble gas configurations.
- Chlorine has a more negative electron gain enthalpy than Fluorine ($Cl > F$) because Fluorine's small size leads to significant inter-electronic repulsion in its dense 2p subshell.

Electronegativity (EN):

The qualitative tendency of an atom in a chemical compound to attract shared pairs of electrons towards itself.

- Fluorine is the most electronegative element (EN = 4.0 on Pauling Scale).
- Trend: Increases across a period, decreases down a group.

16. Biology: Body Fluids and Circulation Essentials

Blood is a special connective tissue consisting of a fluid matrix, plasma, and formed elements.

Composition of Blood Plasma (55% of Blood Volume):

- Water (90-92%), Proteins (6-8%: Fibrinogen, Globulins, Albumins).
- **Fibrinogen**: Needed for clotting. **Globulins**: Involved in defense mechanisms. **Albumins**: Maintain osmotic balance.

Formed Elements (45% of Blood Volume):

1. **Erythrocytes (RBCs)**: Count 5-5.5 million/mm³. Formed in red bone marrow. Contain Hemoglobin. Lifespan: 120 days. Destroyed in the spleen (graveyard of RBCs).
2. **Leukocytes (WBCs)**: Nucleated, count 6000-8000/mm³. Divided into Granulocytes (Neutrophils, Eosinophils, Basophils) and Agranulocytes (Lymphocytes, Monocytes).
3. **Thrombocytes (Blood Platelets)**: Count 150,000-350,000/mm³. Produced from Megakaryocytes. Crucial for blood coagulation.

17. Blood Grouping and Clotting Mechanism

ABO Blood Grouping System:

Based on the presence or absence of two surface antigens on RBCs: Antigen A and Antigen B.

- **O Negative:** Universal Donor (lacks A, B, and Rh antigens).
- **AB Positive:** Universal Recipient (contains A, B, and Rh antigens, lacks antibodies).

Mechanism of Blood Coagulation (Clotting):

An injury stimulates platelets to release certain coagulation factors, initiating a cascade mechanism:

1. Thromboplastin Complex (Thrombokinase) is formed.
2. Prothrombin -----[Thrombokinase + Ca^{2+}]-----> Thrombin
3. Fibrinogen -----[Thrombin]-----> Fibrin (clot mesh)

Calcium ions (Ca^{2+}) play an essential role in all regulatory steps of the clotting cascade.

18. Human Circulatory System & Heart Architecture

The human heart is a muscular, myogenic, 4-chambered organ derived from the embryonic **mesoderm**.

Internal Structure Components:

- **Atria:** Upper thin-walled receiving chambers. Separated by thin inter-atrial septum.
- **Ventricles:** Lower thick-walled pumping chambers. Left ventricle wall is three times thicker than right ventricle to support systemic pressures.
- **Valves:** Ensure unidirectional blood flow without backflow:
 - Tricuspid Valve: Positioned between right atrium and right ventricle.
 - Bicuspid (Mitral) Valve: Positioned between left atrium and left ventricle.
 - Semilunar Valves: Guard the openings of the pulmonary artery and systemic aorta.

19. Cardiac Cycle Dynamics and ECG

The sequential events occurring in the heart during one single heartbeat is called a Cardiac Cycle.
Duration: **0.8 seconds** (at 75 beats/min).

Stroke Volume = 70 mL | Cardiac Output = Stroke Volume × Heart Rate ≈ 5000 mL/min

Electrocardiogram (ECG) Waves Breakdown:

- **P-Wave:** Represents electrical depolarization of the atria (leads to atrial contraction).
- **QRS Complex:** Represents depolarization of the ventricles, initiating ventricular contraction.
- **T-Wave:** Represents ventricular repolarization (return from excited to normal relaxed state).

20. Excretory Products and Modes of Excretion

Animals accumulate nitrogenous metabolic wastes that must be cleared out continuously to prevent severe toxicity.

Modes of Excretion across Animal Kingdom:

1. **Ammonotelism:** Excretion of Ammonia. Highly toxic, requires large amounts of water. Examples: Bony fishes, aquatic amphibians, aquatic insects.
2. **Ureotelism:** Excretion of Urea. Less toxic, conserves water. Examples: Mammals, terrestrial amphibians, marine fishes.
3. **Uricotelism:** Excretion of Uric Acid. Least toxic, eliminated as a thick paste or pellet with minimal water loss. Examples: Reptiles, birds, land snails, insects.

Excretory Structures:

- Protonephridia (Flame cells) in Platyhelminthes.
- Nephridia in Annelids. Malpighian tubules in Insects. Antennal/Green glands in Crustaceans.

21. Human Excretory System & Structural Unit

The human excretory system consists of a pair of kidneys, a pair of ureters, a urinary bladder, and a urethra.

The Nephron:

Each kidney contains roughly one million functional units called nephrons. A nephron has two primary structural components:

- **Glomerulus:** A tuft of capillaries formed by the afferent arteriole.
- **Renal Tubule:** Begins with a double-walled cup-like structure called **Bowman's Capsule**, which encloses the glomerulus. Together they form the Malpighian Body or Renal Corpuscle.

The tubule continues sequentially into the Proximal Convoluted Tubule (PCT), Henle's Loop (hairpin shape with descending and ascending limbs), and Distal Convoluted Tubule (DCT), terminating into the Collecting Duct.

22. Physiology of Urine Formation

Urine formation involves three main physical processes occurring along the length of the nephron:

1. **Glomerular Filtration:** Occurs through 3 filtration barrier layers. Normal Glomerular Filtration Rate (**GFR**) = **125 mL/min** (or 180 Liters per day!). Over 99% of this filtrate is reabsorbed.
2. **Reabsorption:**
 - **PCT:** The main site of reabsorption. Nearly 70-80% of electrolytes and water, and 100% of nutrients (glucose, amino acids) are actively reabsorbed here.
 - **Loop of Henle:** Minimum reabsorption occurs here. It plays an essential role in maintaining high osmolarity of medullary interstitial fluid.
3. **Secretion:** Active tubular secretion of **H⁺**, **K⁺**, and ammonia into the lumen to regulate ionic balance and body pH.

23. Concentration of the Filtrate & Renal Regulation

Counter-Current Mechanism:

Mammals possess the unique ability to produce highly concentrated urine using a counter-current multiplier system operating between the Loop of Henle and the surrounding vasa recta capillaries.

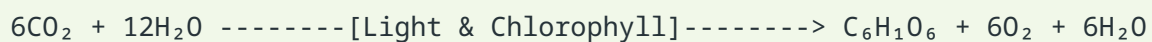
Flow of filtrate in the two limbs of Henle's loop is in opposite directions, creating a steep gradient increasing from **300 mOsmol/L** in the cortex up to **1200 mOsmol/L** deep in the inner medulla. This steep gradient allows smooth osmotic abstraction of water from the collecting ducts to concentrate urine up to 4 times the initial plasma concentration.

Hormonal Regulation:

- **ADH (Vasopressin):** Released by posterior pituitary when osmoreceptors detect water loss. It acts on late DCT/Collecting duct to increase water reabsorption.
- **RAAS System:** Drop in GFR triggers Juxtaglomerular (JG) cells to release **Renin**, converting Angiotensinogen to Angiotensin I and II. Angiotensin II triggers Aldosterone release from adrenal cortex, increasing **Na⁺** and water reabsorption.

24. Photosynthesis in Higher Plants: Fundamental Principles

Photosynthesis is an anabolic, endergonic physico-chemical process by which green plants utilize solar energy to synthesize organic compounds from **CO₂** and water.



Site of Photosynthesis:

Occurs inside specialized double-membrane organelles called **Chloroplasts**.

- **Grana Thylakoids:** Membrane system responsible for trapping light energy and synthesizing ATP and NADPH. (Site of **Light Reactions**).
- **Stroma:** Fluid matrix where enzymatic dark reactions fix gaseous carbon dioxide into sugars. (Site of **Dark Reactions / Calvin Cycle**).

25. The Light Reaction: Photophosphorylation Mechanisms

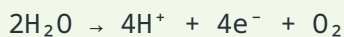
Light absorption, water splitting, oxygen release, and formation of high-energy chemical intermediates (ATP and NADPH) characterize the light reactions phase.

Photosystems (PS I and PS II):

Harvesting complexes containing hundreds of pigments. PS I has a reaction center chlorophyll-a absorbing at **700 nm** (P700). PS II reacts maximally at **680 nm** (P680).

Non-Cyclic Photophosphorylation (Z-Scheme):

Both photosystems operate in series. Electrons expelled from PS II pass down an electron transport chain to PS I. Simultaneously, electrons from PS I are transferred to **NADP⁺** to form NADPH. Electrons lost from PS II are continuously replenished by the **Splitting of Water** (Oxygen-Evolving Complex linked to PS II facing inner thylakoid space):



Cyclic Photophosphorylation:

Operates when only light wavelengths beyond 680 nm are available, or when only PS I functions. Electrons cycle back through the transport loop, generating **ATP only** (No NADPH or **O₂** produced).

26. Chemiosmosis and Carbon Fixation (C3 Cycle)

Chemiosmotic Hypothesis (Peter Mitchell):

ATP synthesis is driven by a proton gradient established across the thylakoid membrane. Protons accumulate inside the thylakoid lumen due to water splitting and the cytochrome transport mechanism. This gradient breaks down as protons diffuse back to the stroma through the **F₀-F₁ ATP Synthase** enzyme complex, driving ATP synthesis.

The Calvin Cycle (C3 Pathway):

Operates in all photosynthetic plants. It has three consecutive biochemical stages:

1. **Carboxylation:** Fixation of **CO₂** into a stable 3-carbon compound (3-PGA) catalyzed by the abundant enzyme **RuBisCO**. Primary acceptor: RuBP (5-carbon sugar).

2. **Reduction:** Enzymatic steps consuming 2 ATP and 2 NADPH molecules per fixed CO_2 to produce triose phosphates.
3. **Regeneration:** Regeneration of primary acceptor RuBP, requiring 1 additional ATP molecule.

Net Energy Cost for 1 Glucose Molecule (6 Turns of C3 Cycle): 18 ATP + 12 NADPH

27. C4 Pathway and Photorespiration (C2 Cycle)

Photorespiration (C2 Cycle):

A wasteful process occurring in C3 plants when RuBisCO binds to Oxygen instead of CO_2 under high temperatures and high oxygen concentrations. It results in no synthesis of ATP, NADPH, or sugars, while releasing previously fixed carbon dioxide.

C4 Pathway (Hatch-Slack Pathway):

An adaptation in tropical plants (e.g., Maize, Sugarcane) to bypass photorespiration entirely. These plants exhibit specialized internal anatomy known as **Kranz Anatomy** (dimorphic chloroplasts in mesophyll and bundle sheath cells).

- Primary CO_2 acceptor in mesophyll: PEP (Phosphoenolpyruvate).
- Primary fixing enzyme: PEP carboxylase (PEPcase), which is insensitive to oxygen.
- First stable product: Oxaloacetic Acid (OAA), a 4-carbon organic acid.
- Bundle sheath cells possess dense RuBisCO to run the normal Calvin cycle under localized high concentrations of CO_2 .

28. Plant Growth Regulators (PGR) - Auxins & Gibberellins

Plant growth regulators are small, simple chemical molecules secreted in low quantities that modulate growth, differentiation, and development phases in plants.

1. Auxins (Indole compounds - e.g., IAA, IBA, NAA, 2,4-D):

- Discovered initially by Charles Darwin and Francis Darwin; isolated by F.W. Went from oat coleoptiles.

- Key Functions: Promote apical dominance (inhibit lateral buds), initiate rooting in stem cuttings, prevent premature fruit drop, and act as selective weedicides (2,4-D clears dicot weeds).

2. Gibberellins (Terpenes - e.g., GA₁, GA₂, GA₃):

- Discovered in Japan tied to the 'Bakanae' (foolish seedling) disease of rice caused by the fungus *Gibberella fujikuroi*.
- Key Functions: Promote rapid internodal elongation (bolting in rosette plants like cabbage), increase length of grape stalks, delay senescence, and speed up the malting process in brewing industries.

29. Cytokinins, Ethylene, and Absciscic Acid (ABA)

3. Cytokinins (Adenine derivatives - e.g., Kinetin, Zeatin):

- Discovered by Skoog and Miller as a growth factor promoting cell division from herring sperm DNA.
- Key Functions: Promote active cytokinesis (cell division), delay leaf senescence (Richmond-Lang effect), overcome apical dominance, and promote lateral shoot growth.

4. Ethylene (Gaseous Regulator - C₂H₄):

- Key Functions: Speeds up commercial fruit ripening, induces horizontal growth of seedlings, breaks seed and bud dormancy, promotes root hair formation, and induces flowering in pineapples.

5. Absciscic Acid (Stress Hormone - ABA):

- Acts as a general plant growth inhibitor and antagonist to Gibberellins (GA).
- Key Functions: Stimulates closure of stomata during severe water stress, induces seed dormancy to survive unfavorable periods, and promotes abscission of leaves and fruits.

30. Photoperiodism, Vernalization, and Seed Dormancy

Photoperiodism:

The developmental response of plants to the relative duration of light and dark periods to initiate flowering.

- Short-Day Plants (SDP): Require exposure to light below a critical photoperiod.
- Long-Day Plants (LDP): Require exposure to light exceeding a critical duration.

- Day-Neutral Plants (DNP): Flowering response is completely independent of photoperiod length.

Vernalization:

The chilling treatment or dependence of certain plants on exposure to low temperatures to quantitatively or qualitatively promote subsequent flowering.

Seed Dormancy:

A state of temporary metabolic arrest in seeds driven by endogenous inhibitors like ABA, phenolic acids, or immature embryos, preventing germination until optimal environmental conditions are fully met.